

Book I. Prop. XXVI. Theor. *If two triangles have two angles of the one respectively equal to two angles of the other ($\triangle = \triangle$ and $\triangle = \triangle$), and a side of the one equal to a side of the other similarly placed with respect to the equal angles, the remaining sides and angles are respectively equal to one another. Case I.*

Let --- and --- which lie between the equal angles be equal, then

$$\text{---} = \text{---}.$$

For if it be possible, let one of them --- be greater; make $\text{---} = \text{---}$, draw --- .

In $\triangle$ and $\triangle$ we have

$$\text{---} = \text{---}, \triangle = \triangle, \\ \text{---} = \text{---};$$

$$\therefore \triangle = \triangle \text{ (pr. I.4)}$$

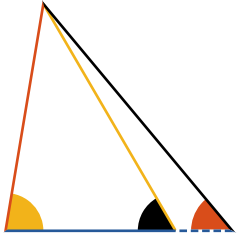
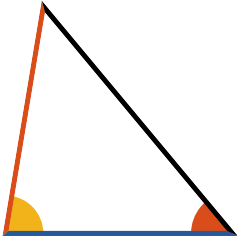
$$\text{but } \triangle = \triangle \text{ (hyp.)}$$

and therefore $\triangle = \triangle$, which is absurd;

hence neither of the sides --- and --- is greater than the other;

and $\therefore$ they are equal;

$$\therefore \text{---} = \text{---}, \text{ and } \triangle = \triangle \text{ (pr. I.4).}$$



Book I. Prop. XXVI. Theor. (cont.)
Case II.

Again, let  = , which lie opposite the equal angles  and .

If it be possible, let  > , then take  = , draw .

Then in  and  we have  = ,  =  and  = ,

$\therefore$  =  (pr. I.4)

but  =  (hyp.)

$\therefore$  =  which is absurd (pr. I.16).

Consequently, neither of the sides  or  is greater than the other, hence they must be equal. It follows (pr. I.4) that the triangles are equal in all respects.

Q. E. D.